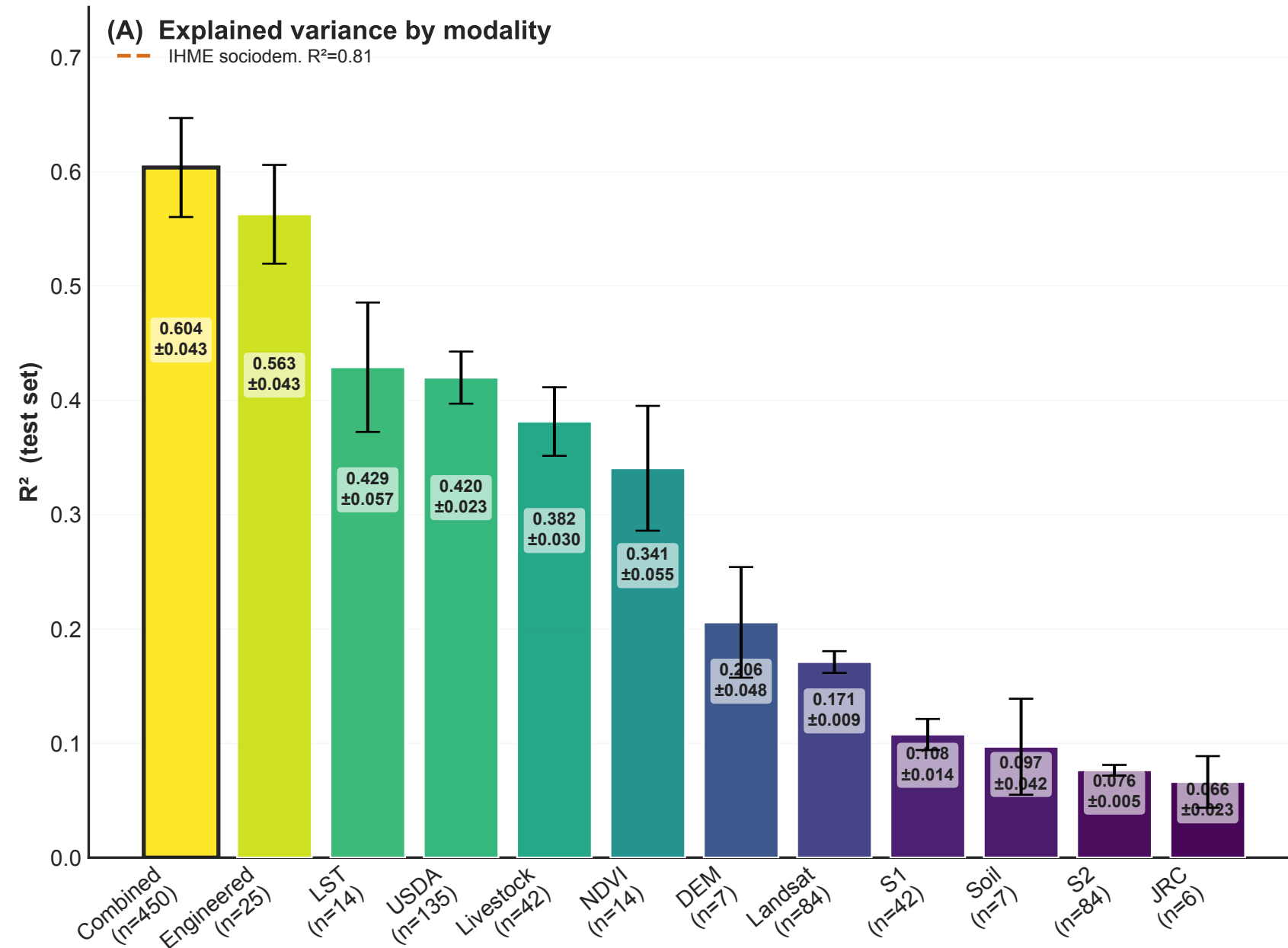


Model Modalities: Master Summary ($R^2 \pm SD$ & MAE $\pm SD$)

Multimodal fusion markedly outperforms single-modality alternatives

(A) Explained variance by modality

--- IHME sociodem. $R^2=0.81$



(B) Mean Absolute Error by modality

Modality	RMSE
Combined	1.55
Engineered	1.63
LST	1.86
USDA	1.87
Livestock	1.94
NDVI	2.00
DEM	2.19
Landsat	2.24
S1	2.33
Soil	2.34
S2	2.37
JRC	2.38

